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An Anomaly-Enhanced Digital Twin (AEDT) Model for Intelligent Inspection and Management of Civil Infrastructure

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ABSTRACT

The increasing frequency of extreme climate events poses significant risks to slope infrastructure, while traditional inspection methods are often inefficient and unsafe. Although unmanned aerial vehicles (UAVs) combined with structure-from-motion (SfM) provide high-fidelity 3D models, they lack the semantic understanding necessary for automated damage assessment. This study addresses this gap by developing and validating an anomaly-enhanced digital twin (AEDT) framework. The proposed system integrates multiview UAV imagery, SfM-based 3D reconstruction, and a convolutional neural network (CNN) for automated anomaly classification. This information is then fused into an interactive, geographic information system (GIS)-compatible DT platform for lifecycle management. A case study on a soil and water conservation (SWC) structure in central Taiwan was conducted for verification. The deep learning module achieved a macroaverage F1-score of 0.81, demonstrating balanced performance across erosion, spalling, siltation, and collapse classes. This was validated on a held-out test set derived from a total of 2000 annotated images spanning four anomaly types with three severity levels. Furthermore, the AEDT-based workflow reduced on-site inspection time by ~63% compared to conventional manual methods. The resulting AEDT model provides a dynamic, semantically enriched 3D representation of the infrastructure, linking geometric data with damage attributes and historical maintenance records. This research demonstrates a feasible and scalable solution for intelligent infrastructure monitoring, offering a robust tool for enhancing climate resilience and enabling proactive asset management.

1 | Introduction

The frequency and intensity of extreme climate events have increased significantly in recent decades, posing escalating risks to soil engineering and civil infrastructure systems. Extreme precipitation, prolonged heatwaves, and intensified typhoons not only accelerate the physical degradation of infrastructure components—such as surface erosion, cracking, and material fatigue—but also destabilize geotechnical systems through hydrological fluctuations, increased pore water pressure, and slope saturation.

Soil and water conservation (SWC) structures, which play a critical role in mitigating sediment transport, maintaining slope stability, and protecting downstream communities, are particularly vulnerable. This vulnerability is compounded in mountainous or remote terrains where SWC installations are often located. In such areas, routine inspection and maintenance are hindered by inaccessibility, complex topography, and unpredictable weather patterns. As a result, latent damage may go undetected, increasing the likelihood of progressive deterioration and sudden failure.

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These growing challenges necessitate a paradigm shift in monitoring practices—moving from periodic, manual inspection methods toward intelligent, automated, and spatially contextualized solutions capable of operating in dynamic and data-scarce environments. To address this need, researchers are increasingly using unmanned aerial vehicles (UAVs) equipped with high-resolution sensors to conduct infrastructure inspections. This technology has become an important alternative to traditional ground surveys due to its high efficiency, safety in accessing hard-to-reach areas, and ability to obtain high-precision data.

However, simply collecting large amounts of image data is not enough in itself; the potential of the raw data needs to be fully unleashed through advanced processing techniques. To this end, UAV photogrammetry combined with structure-from-motion (SfM) technology can quickly and noncontact produce dense 3D point clouds that accurately represent the surface morphology. The advantage of such reconstructions is that they can provide high-fidelity geometric models, which are particularly useful for quantitatively recording erosion, deformation, and geometric anomalies. However, the main limitation of this method is that its output is purely geometric in nature and lacks semantic understanding. In other words, although the point cloud accurately depicts the "shape" of the structure, it cannot automatically identify whether a feature is a "crack," "spalling," or a harmless shadow. This need for semantic annotation of massive visual data provides an entry point for the next stage of technology application.

Simultaneously, advances in computer vision, particularly the application of deep convolutional neural networks (CNNs)—have greatly improved the accuracy and automation of structural defect detection. CNN–based models have demonstrated significant potential and high precision in detecting surface cracks and other forms of damage. Emerging methods further combine object detection with generative networks, improving robustness and speed even under varying lighting and texture conditions.

Despite these advances in technology, current inspection systems often lack semantic integration and temporal continuity. UAV–CNN pipelines typically operate in a 2D image space and are not embedded in a spatially contextualized framework that supports lifecycle monitoring or predictive analytics. Digital twin (DT) technology offers a promising solution by creating a dynamic 3D representation of infrastructure linked to sensor data and maintenance records. In geotechnical applications, DTs are increasingly being used for real-time risk prediction and slope deformation monitoring through data fusion and artificial intelligence (AI)-driven analytics. However, implementations tailored for SWC structures remain limited and often lack the integration of semantic defect recognition and geometry-aware modeling.

This study addresses this gap by proposing an integrated anomaly-enhanced DT (AEDT) framework that combines UAV-SfM-based 3D modeling, CNN–based semantic anomaly detection, and rule-based degradation classification in a geographic information system (GIS)-compatible environment. The goal is to build a scalable smart system for time-series inspection, condition assessment, and preventive maintenance of slope infrastructure to enhance its resilience under changing climate conditions. By connecting geometric fidelity and semantic intelligence into a dynamic, inspectable, and updatable DT, this study advances the field of smart SWC infrastructure monitoring and climate-adaptive resilience.

2 | Literature Review

The increasing demand for automation in infrastructure inspection has fueled growing research interest in integrating remote sensing, AI, and digital modeling technologies. For SWC structures—often located in remote and rugged areas—manual inspection presents significant limitations. As a result, there is a growing need for efficient, objective, and scalable alternatives to traditional practices. Recent developments emphasize a multidisciplinary convergence, combining UAV photogrammetry, deep learning algorithms, and DT frameworks to address these challenges in a more intelligent, adaptive, and holistic manner.

2.1 | Deep Learning for Structural Anomaly Detection

With the rapid advancement of computer vision, deep learning has emerged as a pivotal tool for defect detection. Foundational research by Cha et al. [1] first demonstrated the efficacy of CNNs for robust crack detection, significantly outperforming traditional edge detection methods. Building on this, Cha et al. [2] further advanced the field by implementing faster region-based CNNs (Faster R-CNN) for the simultaneous detection of multiple damage types, establishing a benchmark for automated visual inspection.

In surface defect inspection, Yang et al. [3] proposed a deep learning-based system that utilizes multiscale and channel-compressed features, combining multiple convolutional layers with varied kernel sizes to better accommodate cluttered backgrounds and variably sized defects. For cement concrete bridge surfaces, Zhu and Song [4] developed an intelligent classification model by modifying the classic VGG-16 network, which achieved superior accuracy in distinguishing seven distinct defect categories compared to shallower networks. Similarly, for underground parking structures, Shin et al. [5] utilized a Faster R-CNN model to objectively locate and classify common defects such as paint peeling and leakage from a large image dataset.

The application of deep learning extends to specialized infrastructure monitoring. For railway tracks, Dang et al. [6] introduced an efficient image classification model that leverages a CNN to analyze RGB spectral images of ultrasonic signals, enabling high-accuracy condition identification for large-scale monitoring. In water distribution systems, Shukla and Piratla [7] developed a deep learning algorithm using a modified AlexNet CNN to detect pipeline leakages from vibration signal images, reaching up to 98% accuracy. For internal defects in reinforced concrete, Kuchipudi and Ghosh [8] presented a region-based CNN model that employs RoIAlign on ultrasonic images to automatically detect and segment flaws, attaining a mean average precision (mAP) as high as 0.98.

In manufacturing, Meister and Wermes [9] evaluated the performance of CNN– and RNN–based algorithms for analyzing laser scanner data, demonstrating that a combined CNN–RNN architecture is highly effective for real-time defect classification in automated fiber placement processes. To provide a broader perspective, review articles have also been pivotal. Ming et al. [10] conducted a literature review of deep CNN applications for defect detection in 3C products, summarizing various frameworks and analyzing their respective challenges. Likewise, Ali et al. [11] published a comprehensive review on the implementation of CNNs for crack detection in civil structures, detailing all procedural

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aspects and highlighting the advantages over conventional methods.

Collectively, these studies demonstrate the immense potential of deep learning, especially CNNs and their variants, to overcome the limitations of traditional methods in terms of time, cost, and accuracy for defect detection across a wide range of materials and structures.

2.2 | Application of UAVs in Structural Damage Assessment and Monitoring

UAVs have become indispensable tools in inspecting difficult-to-access civil structures such as slopes, bridges, and postdisaster zones. The operational health inspection and postdisaster damage assessment of civil structures have long been critical challenges in both research and practice. Traditional visual inspection methods suffer from high variability, subjectivity, and the substantial cost and time required for large-scale infrastructure, leading to insufficient temporal resolution for effective decision-making. In recent years, the rapid development and maturation of UAV technology has presented new opportunities to address these challenges. With their advantages of high spatial resolution, capacity to generate stereoscopic imagery and its derivatives, and platform flexibility, UAVs have opened numerous new possibilities in the field of damage mapping. This review aims to synthesize recent key research on the use of UAVs for structural damage detection, quantification, and monitoring, providing a comprehensive perspective on the progress and potential within this field.

A critical evolution in this domain is the shift from manual piloting to fully autonomous systems. Kang and Cha [12] developed an autonomous UAV system utilizing ultrasonic beacons for structural health monitoring (SHM) in GPS-denied environments, overcoming major navigation limitations in confined spaces. Furthermore, Ali et al. [13] introduced a real-time multiple damage mapping framework using autonomous UAVs, while recent innovations have integrated obstacle-avoidance capabilities with fiducial marker-based localization to ensure safe operations in complex structural geometries [14]. These advancements highlight the growing feasibility of automated data acquisition in complex structural environments. Pan et al. [15] integrated UAV photogrammetry with panoramic imaging and fuzzy semantic evaluation, enhancing both spatial coverage and automated condition assessment for SWC infrastructure. Kerle et al. [16] reviewed the evolution of UAV applications in structural disaster damage mapping, charting the progression from providing simple descriptive overviews of disaster scenes to employing more complex texture and segmentation-based methods, and ultimately to utilizing advanced deep learning approaches with multitemporal and multiview imagery for comprehensive damage characterization. Their review also explored emerging developments such as the practicality of UAV mapping strategies for first responders, improvements in machine learning, and the application of UAVs in GPS-denied environments and infrastructure maintenance.

For specific damage assessment of infrastructure like bridges, several studies have demonstrated the potential of integrating UAVs with advanced image processing and AI. Ellenberg et al. [17] demonstrated the use of computational algorithms with UAV-acquired imagery for quantitative damage assessment, employing a postprocessing framework involving homography and lens distortion correction to extract precise measurements of deformation, change detection, and crack patterns. Peng et al. [18] developed a UAV-based machine vision method using hybrid feature learning for bridge crack identification and width quantification, which integrated an R-FCN network with Haar-AdaBoost to achieve over 90% accuracy in real-world crack measurements. Kao et al. [19] combined the YOLOv4 deep learning model with UAV image processing techniques to extract and quantify bridge cracks, achieving 92% accuracy and a width measurement precision of up to 0.22 mm by converting identified crack images to binary maps for precise edge detection.

Beyond static damage assessment, UAVs are also being utilized for dynamic response monitoring. Wang et al. [20] proposed a UAV-based video analysis methodology to track earthquake-induced building displacements. By correcting sensor motion and tracking natural features at the pixel level, their method extracts dynamic responses with engineering-grade accuracy (RMSE <2 cm), as validated by shake-table tests.

The application of UAVs has further expanded to more specialized and efficient infrastructure inspections. Benelmostafa and Medromi [21] introduced a multitask YOLO model, named PowerLine-MTYOLO, for the simultaneous segmentation of power cables and detection of broken strands from UAV imagery. By integrating novel modules into a YOLO v8-based architecture, their model significantly improved detection and segmentation performance, offering a promising solution for autonomous aerial inspections. Collectively, these studies demonstrate that UAVs serve as a powerful tool, significantly enhancing the efficiency and precision of structural damage assessment and infrastructure maintenance, evolving capabilities from qualitative descriptions to precise quantitative analysis, dynamic monitoring, and complex multitask processing.

2.3 | DTs for Infrastructure Maintenance Management

The concept of DT, initially developed by NASA for spacecraft simulation, has been increasingly applied in civil infrastructure, particularly since 2020. As three-dimensional digital models assume an increasingly vital role in domains such as smart cities, DTs, autonomous driving, and industrial production, the associated data processing, model construction, and applications have become central to current research. Recent literature reveals numerous innovative findings, from the precise semantic segmentation of 3D point cloud data to the application of DTs that integrate AI and extended reality (XR) technologies.

In the realm of 3D point cloud semantic segmentation, Ballouch et al. [22] developed a novel prior-level fusion approach for large-scale urban environments that integrates optical image classifications as prior knowledge into a RandLaNet deep learning model, achieving a 96% intersection over union (IoU). Addressing similar challenges, Bai et al. [23] introduced MFFTNet, a new 3D point cloud segmentation network that leverages a multiscale feature fusion and transformer architecture to enhance performance by globally modeling overall shape while embedding local details. For detailed urban modeling, Münzinger et al. [24] proposed a GIS workflow to generate detailed 3D tree models from LiDAR point clouds by classifying the urban forest and parametrically

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reconstructing individual tree crowns, thereby filling a critical information gap in high-resolution urban forest structure.

Within the scope of DT applications, diverse solutions have been presented. Xu et al. [25] developed a DT health monitoring information model for building structures by fusing a building information model (BIM) with a real-scene 3D model, enabling streamlined sensor data management for SHM. Extending to industrial contexts, Chatterjee et al. [26] investigated how immersive collaborative business processes and XR-driven industrial metaverse technologies can facilitate economic value cocreation within 3D DT factories, highlighting the role of interconnected AI and IoT systems. Furthermore, Hu et al. [27] proposed a generative AI-enabled facility layout design paradigm that integrates semantic models, such as the Asset Administration Shell (AAS), with knowledge graphs and large language models (LLMs) to automate and optimize layout generation from contextual prompts, aligning with Industry 4.0 principles.

Collectively, these research efforts advance the application of 3D digital models in smart cities, industrial production, and building management. Specifically, by combining deep learning, data fusion, AI, and DT technologies, they significantly enhance data processing precision, system intelligence, and real-time monitoring capabilities. While photogrammetry and SfM remain industry standards for large-scale modeling, emerging technologies such as neural radiance fields (NeRF) are redefining the boundaries of 3D reconstruction. Recent studies by Kim and Cha [28, 29] demonstrated Nerfacto-based approaches capable of pixel-wise damage mapping with superior visual fidelity. However, current DT applications in civil engineering often prioritize interoperability with existing GIS and BIM standards. Consequently, this study adopts SfM-based explicit geometric modeling to ensure seamless integration with legacy asset management systems.

2.4 | Research Gap and Proposed Solution

To position our contributions against prior art, we summarize representative studies and contrast integration level, temporal capability, and novelty in Table 1.

In summary, although UAVs and AI have proven effective for enhancing infrastructure inspection, and DTs provide a robust framework for lifecycle asset management, current applications often operate in silos. Few existing systems fully integrate UAV-acquired imagery, AI-based anomaly detection, and semantically enriched 3D DT models into a unified, scalable platform. This lack of convergence creates a significant research gap in critical areas such as multimodal data integration (e.g., 2D imagery, 3D geometric reconstructions, and semantic annotations), automated anomaly interpretation, and the embedding of inspection results into an interactive digital environment that facilitates decision-making. This fragmentation results in missed opportunities for temporal tracking, contextual interpretation, and predictive analytics, all of which are essential for proactive infrastructure management. These limitations are particularly acute for SWC infrastructure, where challenging topographies and dispersed locations exacerbate inspection constraints.

This study addresses these gaps through the development of the AEDT framework, which uniquely integrates: (1) multiview UAV photogrammetry for comprehensive geometric coverage; (2) deep learning-based semantic classification of surface anomalies for enhanced diagnostic accuracy; and (3) level of detail 3 (LOD3) DT modeling for spatially and semantically enriched representation. The proposed AEDT model not only links heterogeneous data sources into a coherent model but also supports spatiotemporal inspection analysis and dynamic anomaly tracking, enabling intelligent, context-aware decision-making tailored to geotechnical infrastructure in subtropical, hazard-prone regions. In doing so, this research provides a novel, end-to-end solution that advances the field beyond existing fragmented approaches and lays a foundation for resilient infrastructure monitoring.

3 | Research Methods

This study constructs an AI-assisted inspection framework for artificial slope structures. This framework deeply integrates several advanced technologies to form a complete workflow from data acquisition to decision support. The core technical path is as follows: First, multiview image acquisition from UAVs is used to overcome the limitations of traditional manual inspections in accessibility, safety, and efficiency. Next, SfM algorithms are employed for 3D reconstruction, transforming 2D image sequences into a 3D geometric model with real-world scale. Subsequently, deep learning is introduced for anomaly recognition on the structure's appearance, automating the process of identifying potential deterioration or damage from a vast number of images. Finally, through semantic DT modeling, the geometric model is fused with the identified anomaly information and engineering attribute data. The final output of this process is a semantically annotated AEDT model.

The entire AEDT framework can be conceptualized as a four-layer modular architecture, each corresponding to the technical implementation detailed in Sections 3.1–3.3:

1. The data acquisition layer focuses on collecting multiangle UAV imagery and supplemental on-site data to ensure comprehensive spatial coverage. This step provides the raw data input for subsequent reconstruction workflows (see Section 3.1).
2. The model construction layer processes the captured images using SfM and multiview stereo (MVS) techniques to generate dense 3D point clouds and photorealistic mesh models. These geometric models represent the slope infrastructure

TABLE 1 | Comparison with representative studies.

Reference	Method	Integration level	Temporal	Novelty
Ellenberg et al. [17]	UAV + CV	2D quantification	No	Early quantitative UAV damage assessment
Xu et al. [25]	BIM + 3D Scan (DT)	DT integration	Yes	SHM-oriented DT for buildings
This study (AEDT)	(UAV-SfM) + CNN	Anomaly-enhanced DT	Yes	Geometry-semantics-time fusion for SWC slopes

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with LOD3 accuracy, forming the spatial foundation for semantic integration (also detailed in Section 3.1).

3. The anomaly interpretation layer applies deep CNNs trained on annotated imagery to detect and classify surface anomalies such as erosion, spalling, or collapse. The identified features are then quantified and tagged with semantic labels representing defect type and severity (refer to Section 3.2).
4. TheDT integration layer merges geometric, semantic, and contextual metadata into a GIS-compatible, interactive DT environment. This includes spatial tagging of defects, time-stamped inspection records, and visualization modules for decision support and maintenance planning (as described in Section 3.3).

Together, these layers operationalize the AEDT concept, translating raw UAV imagery into a semantically enriched, update-ready DT that enhances condition monitoring, facilitates anomaly tracking, and enables proactive infrastructure management.

3.1 | SfM-Based 3D Modeling

The 3D modeling process in this study begins with high-quality data acquisition. We utilize a UAV equipped with a high-resolution RGB camera for aerial photography. To ensure the spatial consistency and georeferencing accuracy of the reconstructed model, the flight mission employs automated flight path planning with GPS positioning enabled throughout. During flight planning, we set a 70% forward overlap and a 60% side overlap. Prior to model input, all raw images underwent quality control. Images with significant motion blur (flagged by a perceptual blur detector) or severe over/underexposure were excluded to ensure stable SfM reconstruction and downstream anomaly detection. Flight parameters targeted an average ground sampling distance (GSD) of ~1.5 cm/pixel. This high overlap ratio is crucial for ensuring that the SfM algorithm can find enough tie points to robustly solve for camera poses and scene structure. The flight altitude is maintained at a relative height of ~25–30 m, and shooting angles and intervals are fine-tuned based on the site's topography. In addition to automated flights, we supplement with a handheld camera to capture more detailed textures of key structures or areas obscured by vegetation. After the image data is imported, it is processed using SfM and MVS techniques to generate sparse and dense point clouds. Finally, the point cloud data undergoes meshing and texturing procedures to generate a photorealistic triangular mesh model, which is then converted to the LOD3 format. This preserves the structure's geometry and surface information while ensuring a lightweight model. This step involves feature alignment, camera intrinsic parameter estimation, dense reconstruction, and mesh generation.

Compared to other 3D reconstruction methods such as LiDAR-only or SLAM-based approaches, SfM offers several distinct advantages for this study. LiDAR systems, while highly accurate, are cost-prohibitive and less accessible in complex terrain or remote areas. SLAM-based methods are primarily suitable for indoor or limited-range environments and may struggle with scale estimation in large outdoor areas. SfM, on the other hand, allows for detailed and scalable reconstruction using only passive optical sensors, making it a cost-effective and practical choice for large-scale slope monitoring. Additionally, the integration with UAVs enables coverage of difficult-to-access terrain with minimal risk and logistical complexity.

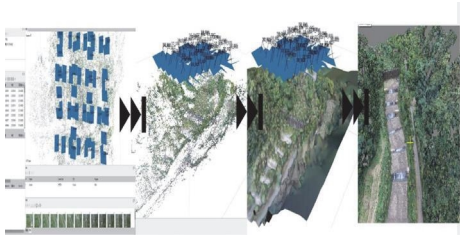


FIGURE 1 | Polygon mesh model diagram.

Process of building a 3D model of the inspection area (Figure 1):

- Step 1: Import image folders: After downloading the large amount of images collected by the UAV from the scene to the computer, they are organized into a dedicated folder and then imported into the image processing software.
- Step 2: Align Photos: Using the software's general and quality settings, this stage finds matching points between overlapping images to calculate the position of each photo, which then allows for the construction of a sparse point cloud model. Testing has shown that using a mid-level image alignment setting reduces processing time and improves the grayscale effect, which facilitates the analysis of differences between images taken at different times.
- Step 3: Build a Dense Cloud: After completing the sparse point cloud model, a build quality is selected in the software. This selection is based on the image positions calculated in the previous step. The program uses the position and depth-of-field information from each camera to perform an information overlay process, combining them into a dense point cloud.
- Step 4: Build Mesh: Following the "Image Alignment" and "Build Dense Cloud" steps, the mesh is generated. This step, along with dense cloud generation, is the most time-consuming but also the most critical part of the process. Therefore, a lower quality setting is used for the initial model build. The reason for using low quality first is that it requires less time, allowing users to quickly review the model for potential errors. After verification, the model is rebuilt at a higher image quality. Once the dense point cloud is finalized, a polygonal mesh model is generated from the dense point cloud data.

3.2 | Deep Learning-Based Anomaly Identification

This study trained a CNN model for surface anomaly identification. The training data consisted of manually annotated images, including various types of anomalies such as erosion, spalling, siltation, and collapse (Figure 2). After calibration, an image recognition model was created, which was then used for identification and verification using the retained images.

The methodology adopted in this study is supervised learning. In this paradigm, the model is provided with input data that has explicit labels (i.e., image patches and their corresponding anomaly classes). During the training process, CNN learns a mapping

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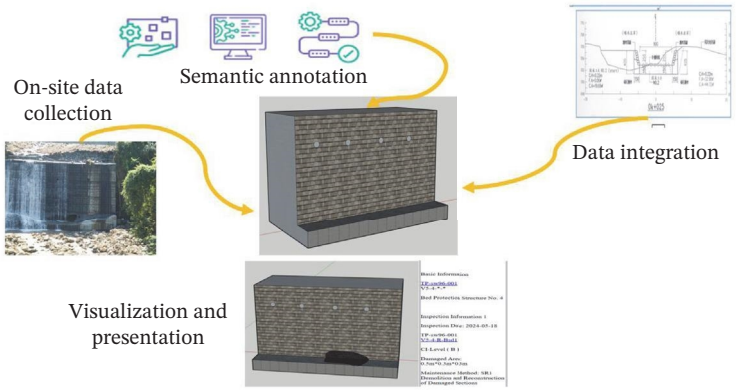


FIGURE 2 | Integration of 3D semantic information in digital twins.

function from input images to output classes. The labeled training images are fed into the model, which performs a forward propagation to get a prediction. This prediction is then compared with the true label to calculate a loss value. Next, the backpropagation algorithm is used to compute gradients, and an optimizer is used to update the network's weights, with the goal of minimizing the loss. This process is repeated iteratively until the model's performance on a validation set converges. The trained recognition model requires validation testing. Therefore, a portion of the data is retained for subsequent model validation and accurate testing. Images not included in the training are then imported for the system to compare against the learned features and output results.

This study employed a CNN based on the MobileNetV2 architecture as implemented by the Teachable Machine platform. We adopted transfer learning with ImageNet-pretrained weights and fine-tuned the entire network for slope-anomaly classification. This platform, through a visual interface, enables rapid creation and export of CNN–based models, suitable for the automated identification and classification of anomaly images. This system models and tests abnormal conditions of SWC structures, such as erosion, spalling, siltation, and collapse. The model's performance was validated using three common performance evaluation metrics: confusion matrix, learning curve, and accuracy per class.

Furthermore, while pixel-wise semantic segmentation (e.g., Mask R-CNN) offers superior granularity for quantifying crack widths, this study employs patch-based classification. This decision is driven by the specific operational objective of "management-level" assessment. The goal is to categorize slope infrastructure into risk levels (Grade A/B/C) to align with government maintenance regulations, for which classification provides a computationally efficient and robust solution without the need for labor-intensive pixel-level annotation. Although YOLOv8 has demonstrated effectiveness in rapid object localization tasks in previous literature, its primary advantage lies in real-time detection speed rather than the pixel-level accuracy critical for subtle surface anomaly identification. Given this study's focus on precise anomaly characterization and relatively limited training data, CNN architecture offers a more suitable balance between accuracy and computational efficiency, making it preferable over YOLOv8 for this specific application scenario. While transformers offer superior performance on large-scale datasets and complex semantic tasks, they require significantly more data and computational resources. YOLOv8, optimized for real-time object detection, emphasizes speed over pixel-level granularity, which is less ideal for identifying subtle surface anomalies in slope structures. EfficientNet achieves strong performance with fewer parameters but involves a more complex architecture not directly supported by the Teachable Machine platform. CNNs, with their relative simplicity and strong capability in image classification, provided a pragmatic and effective solution for this study's dataset and application scope.

The model was trained with batch size 16, learning rate 0.001, and 20 epochs, using the Adam optimizer and categorical cross-entropy loss, with data augmentation (horizontal/vertical flip, $\pm 15°$ rotation, brightness/contrast jitter, and mild zoom) to improve robustness. Data augmentation strategies included horizontal and vertical flipping, rotation ($\pm 15°$), brightness and contrast adjustments, and slight zoom transformations. These augmentations were essential for improving the model's robustness to different environmental conditions and imaging angles. Training was conducted on a workstation equipped with an NVIDIA GeForce RTX 3060 GPU (12 GB VRAM). The total training time was ~2 h.

3.2.1 | Dataset and Annotation

To establish data credibility and enable reproducibility, we report the dataset composition and labeling protocol. Images were split at the structure level to prevent leakage, ensuring that imagery from the same location did not appear in both the training and validation sets. Images were classified with the assistance of domain experts, including SWC technicians and engineers with over 10 years of field experience, following the classification standards in Section 4.4. Interannotator agreement was assessed, achieving a Cohen's κ of 0.87. Discrepancies were adjudicated by a third senior expert (Table 2).

3.3 | Semantic Integration and DT Construction

It is crucial to distinguish between a static 3D model and a functional DT. While 3D reconstruction provides the geometric shell, the "Twin" status in this study is achieved through the semantic integration of AI-detected anomalies, the temporal linkage of historical inspection logs, and the decision-support logic that recommends maintenance actions. Given the current lack of standardized 3D component libraries for SWC facilities and the inadequacy of traditional 2D drawings in representing spatial geometry and degradation status, this study adopts a strategy of constructing the 3D

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TABLE 2 | Dataset distribution by class, severity, and split.

Anomaly category	Severity	Training	Validation	Total
Spalling	A	51	6	57
Spalling	B	65	7	72
Spalling	C	86	9	95
Crack	A	138	15	153
Crack	B	118	13	131
Crack	C	58	7	65
Erosion	A	72	8	80
Erosion	B	102	11	113
Erosion	C	35	4	39
Siltation	A	53	6	59
Siltation	B	78	9	87
Siltation	C	111	12	123
Scouring	B	32	3	35
Scouring	C	116	13	129
Damage	B	88	10	98
Damage	C	148	16	164
Vegetation good (no collapse)	—	94	11	105
Vegetation poor	B	46	5	51
Vegetation poor	C	48	5	53
Collapse	B	77	9	86
Collapse	C	184	21	205

Note: The dataset was divided into two subsets: 1800 training samples and 200 validation samples.

model based on computer-aided design (CAD) as-built drawings. This approach not only enables precise semantic annotation of deteriorated areas but also facilitates the comprehensive integration of information such as dimensions, materials, and spatial relationships into the model. This overcomes the limitations of 2D simulations and significantly enhances the convenience and efficiency of information integration for subsequent applications.

The following is the DT process:

- Image processing and on-site 3D modeling: Use LiDAR scanning or drone aerial photography technology to obtain high-precision three-dimensional spatial information of SWC facilities and generate a digital "as-is" 3D model that reflects the current state of the structure.
- 3D modeling of water conservation facility data: Obtain 2D images or CAD drawings to create a baseline "as-designed" 3D model. This involves generating a digital 3D model of the facility through software, marking the structural material of each part, and displaying specific dimensions. This ensures that the model details capture the intended structural characteristics (such as the state and shape of the dam body), which, combined with on-site inspection data, will become the basis of the DT.
- Data integration and semantic markup: Collect data related to water conservation, including location, size, and function, and integrate this data into the DT. This involves semantically marking the model, linking specific information (such as degradation patterns identified by the deep learning model) with the corresponding parts of the 3D geometry, and making adjustment marks on the model.
- Visualization platform development: Based on the constructed 3D model and data, a visualization platform is developed to allow managers to intuitively view the operational status of the facility. This interactive visualization, using technologies like dynamic web pages, virtual reality (VR), or augmented reality (AR), allows users to interact with the DT and gain a deeper understanding of its various aspects. Relevant data is presented visually, as shown in Figure 2.

4 | System Implementation and Results

4.1 | Visual Design and System Architecture

Visual design significantly influences how users perceive and retain critical information. In this project, 3D visualization is employed to enhance user attention and focus. A well-organized interface ensures that essential inspection data is not overlooked. Notably, dynamic and immersive 3D visual elements aid memory retention and improve user comprehension, which is especially vital in complex infrastructure monitoring tasks. In the digital age, effective

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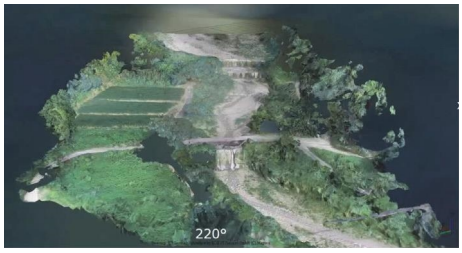


FIGURE 3 | 3D digital layer of inspection area.



FIGURE 4 | 3D digital twin image inspection log information.

information design transcends simple data presentation—DT 3D visualization provides robust tools for constructing responsive, user-centric interfaces.

This system integrates 3D DT visualization of inspection zones with interactive features. Built on Hypertext Preprocessor (PHP), the platform supports dynamic web page generation and real-time 3D model presentations. Users interact with dynamically rendered 3D inspection images, improving information recognition and accessibility. The 3D model viewing interface, created through specialized software, allows users to explore highly detailed digital replicas from multiple angles. These models range from basic geometries to complex terrain and structural elements. By incorporating animation and spatial cues such as parallax scrolling, users can more intuitively explore spatial relationships.

The interface allows step-by-step rotation of model layers, guided by embedded directional markers. Users can rotate models forward or in reverse depending on the desired review perspective. During this process, corresponding inspection records of water conservation facilities are automatically retrieved from the database and displayed visually. This interaction is managed via server-side parameter passing, which triggers retrieval from the backend and returns annotated data to the front-end display (Figure 3).

Furthermore, clickable semantic information points are embedded within the model to show historical maintenance records. These features enable users to review previously taken actions and verify follow-up measures. Administrators can use the backend system to modify or delete information entries, ensuring ongoing data integrity.

In terms of system performance, tests conducted under a simulated multiuser environment show that the average loading time for 3D models (under 50 MB in size) is ~3.2 s, with a 95th percentile response time of 4.5 s. Retrieval of historical data entries averages 1.8 s per query, while interactive actions such as rotation and information point clicking exhibit sub-500 ms latency. The platform has been benchmarked to support up to 50 concurrent users without significant performance degradation, suggesting it is suitable for use in regional-scale monitoring systems.

The system provides differentiated interfaces and access privileges tailored to user roles. Administrators have full control, including the ability to upload datasets, train models, calibrate spatial tags, and modify backend records. Engineers are granted access to upload inspection images, perform spatial alignment with 3D models, and validate AI results (Figure 4). Reviewers from regulatory bodies are provided with a read-only interface with semantic annotations and verification tools for auditing inspection processes and historical maintenance records. This multirole framework ensures operational transparency, data security, and user-specific functionality.

4.2 | Case Management Workflow

This study adopts a five-phase case management process combining deep learning, AI, and spatial mapping to improve inspection efficiency:

- Phase 1: Upload Test Images: Users begin by accessing the new form screen to upload site inspection images. These images are processed using a pretrained deep learning model that classifies visible defects and recommends maintenance actions based on recognized patterns.
- Phase 2: Pattern Interpretation and Form Generation: The system displays AI-detected pattern results alongside a schematic of the DT. This helps users understand each case in spatial and contextual terms. A digital inspection form is automatically generated and archived in the case management system.
- Phase 3: Location Calibration: Users enter the "Add New" interface and spatially locate the relevant inspection zone using the 3D DT. Accurate spatial tagging is achieved through user interaction and knowledge of the site.
- Phase 4: Inspection Review: The platform enables users to visually confirm that each new case is accurately registered. Semantic links are used to validate all information, and any missing or inaccurate data can be identified and corrected. Figure 5 shows the user interface during this step.
- Phase 5: Backend Management: Administrators can edit or delete recorded files through the backend system. The interface ensures streamlined access, batch management of inspection records, and flexible data updates.

4.3 | Case Study and Verification

To demonstrate system functionality, a case study was conducted at the "Shihgang Dam Sediment Control Facility" located in Shihgang District, Taichung City, central Taiwan. The site features a slope-protection system with revetments on both banks.

- Step 1: The AI model identifies deterioration patterns from uploaded images. These patterns are projected onto the 3D model using semantic tagging.
- Step 2: The system compiles a digital report that outlines facility status, proposed remediation strategies, and relevant structural metrics such as dam height and slope angle. Based on site measurements, a 3D facility model is rendered, and visible damage is annotated.
- Step 3: A case link is generated and added to the 3D twin interface. After position calibration, the system automatically registers the three-axis coordinates. Metadata,

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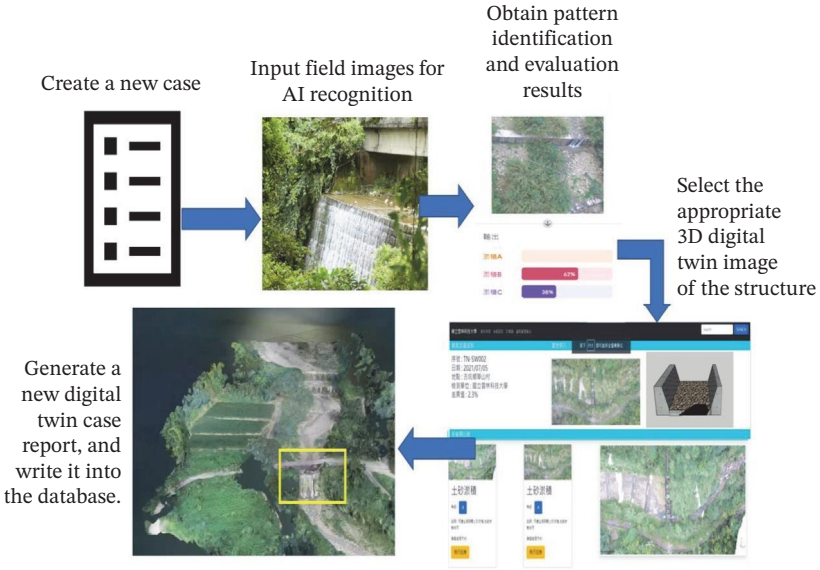


FIGURE 5 | Overall workflow for adding a new case.

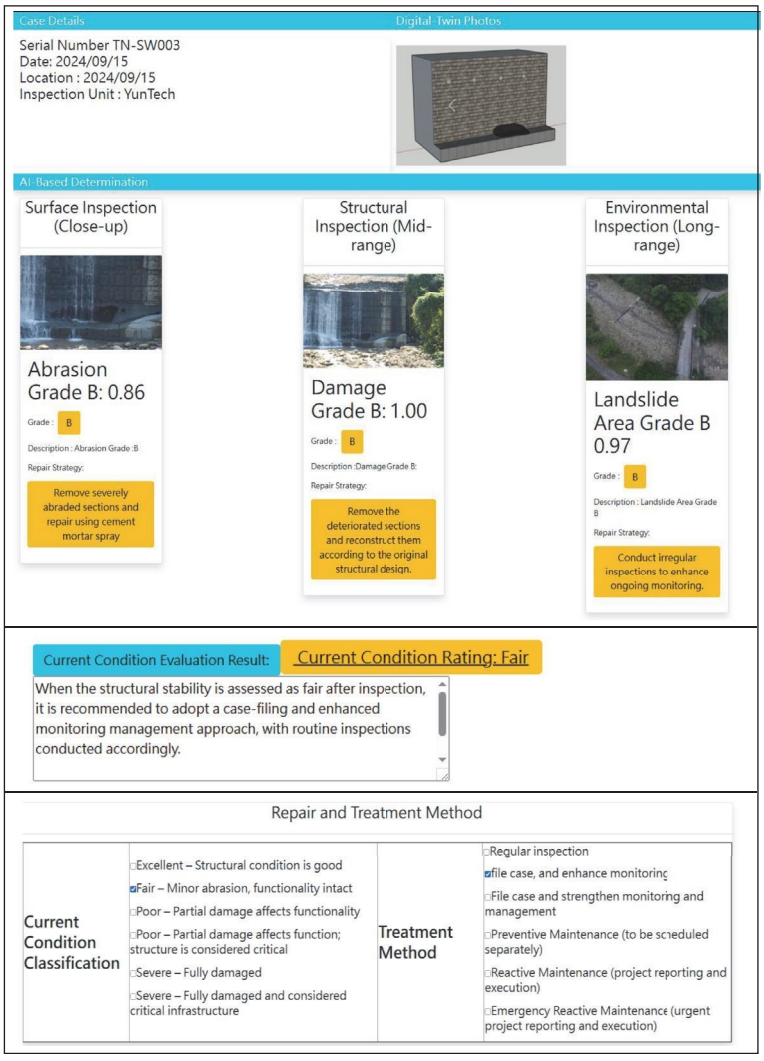


FIGURE 6 | AI recognition rating result interface.

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FIGURE 7 | Information linkage review after case creation.

including image snapshots and textual annotations, is stored in the digital archive. Figures 6 and 7 illustrate these processes.

4.4 | Deep Learning Model Performance Evaluation

To evaluate the performance of the image classification model developed for the automated identification of structural anomalies, this paper summarizes and analyzes three validation metrics: confusion matrix, learning curves for training and testing accuracy, and accuracy per class. First, the learning curves show that the model converges quickly early in training. After approximately the fifth epoch, the training accuracy reaches nearly 100%, while the test accuracy stabilizes at ~81%. This result demonstrates the model's good learning ability and basic generalization. However, a significant gap exists between the training and test curves, which may reflect a slight overfitting phenomenon. To address this and enhance generalization, we implemented optimization strategies such as extensive data augmentation and dropout layers, as shown in the resulting learning curves (Figure 8). After these optimizations, the model achieved a stable overall test accuracy of 81%, demonstrating its potential for practical application.

Analyzing the classification accuracy of each category, the model performed exceptionally well in the erosion-B, spalling-C, and collapse-C categories, achieving accuracy exceeding 0.94. The confusion matrix also showed complete accuracy in the predictions, indicating that these anomaly types possess distinct visual characteristics that the model has learned effectively. While there were a few misclassifications in the medium-to-high accuracy categories, such as erosion-A (0.88), spalling-A (0.83), spalling-B (0.88), and siltation-A (0.83), the model maintained high discrimination capabilities overall. These misclassifications mostly occurred between different levels of the same type, representing an expected level of confusion. However, siltation-B and collapse-B performed relatively poorly, with accuracies of only 0.60. The confusion matrix showed a wide range of predictions, with misclassifications into erosion-A, spalling-B, and collapse-C, potentially due to ambiguous pattern characteristics, high sample heterogeneity, or data imbalance. To address this, ambiguous samples are removed, the number of samples is increased to strengthen the clarity of this category, and a category weight adjustment strategy is introduced into the model

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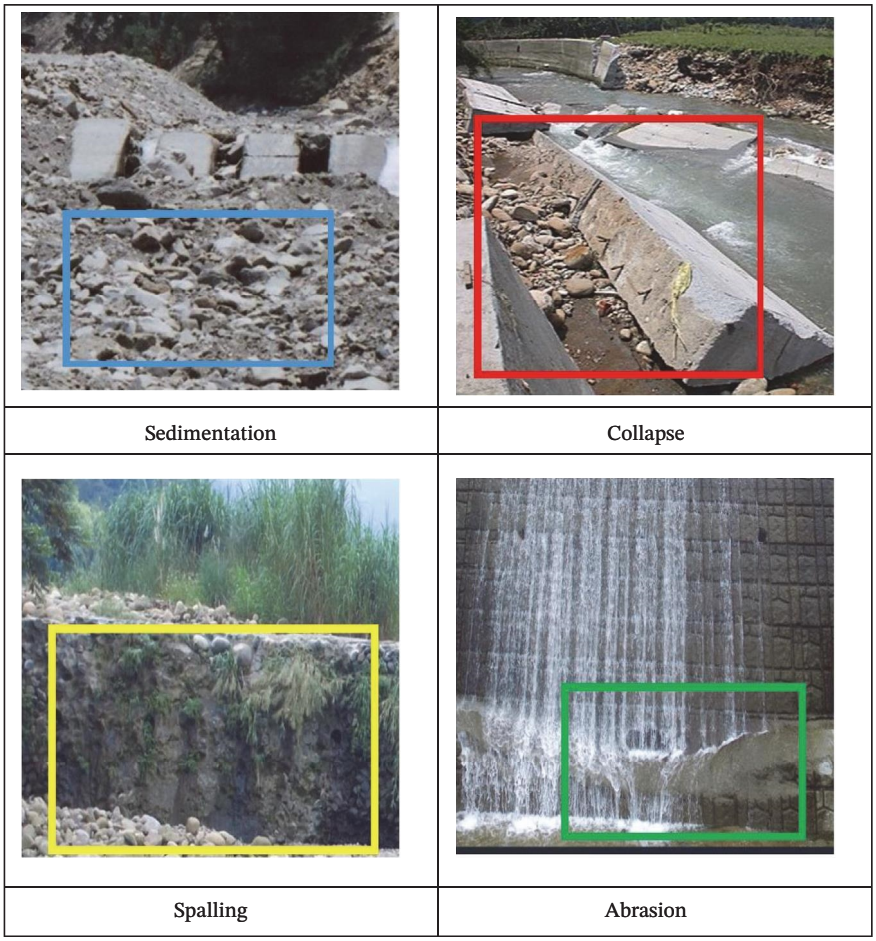


FIGURE 8 | Manually annotated anomalous pattern images.

to improve the prediction accuracy of difficult-to-distinguish categories (Figure 9).

The classification standards for each anomaly type are based on national guidelines such as Taiwan's "Technical Specifications for Soil and Water Conservation" and international references like the ASTM D6433-20 for visual inspection. For example, erosion is classified into levels A–C based on depth and area coverage: Level A (<5 cm depth), Level B (5–15 cm), and Level C (>15 cm or severe surface retreat). Siltation is judged based on the thickness and spread of sediment coverage, while spalling and collapse follow criteria including extent, shape, and risk to structural integrity. Classification decisions are primarily based on visual inspection supplemented by UAV-derived dimensions. Severity levels are determined by using thresholds on physical parameters such as anomaly area, shape complexity, and impact proximity to key structural zones.

A comprehensive analysis shows that this model achieved an overall test accuracy of 81%, with good recognition capabilities for most common anomaly categories, demonstrating its potential for practical application. To address the difficulty in identifying low-accuracy categories and model overfitting, subsequent improvements were made through data augmentation, model architecture optimization, and training strategy adjustments, ultimately improving system stability and reliability. Beyond overall accuracy (81%), we report class-balanced metrics: the model achieved a macroaveraged F1-score of 0.79 and a microaveraged F1-score of 0.81. For the challenging siltation-B class, precision was 0.55 and recall 0.60, reflecting visual similarity with adjacent severity levels. To quantify uncertainty, we performed bootstrap resampling (1000 replicates) on the test set; the 95% confidence interval for the macro-F1 was 0.76–0.82, indicating a stable performance estimate.

4.5 | Integrated AI Reporting and System Effectiveness

This project developed a deep learning-based image recognition architecture tailored to water conservation facilities. Using labeled degradation pattern data, the AI model classifies damage and assesses current conditions. The system employs a contactless condition evaluation process that reduces inspection time and labor requirements. The final output is a semantically enhanced 3D DT with inspection-linked reports. In emergency scenarios, the platform can automatically generate alerts and suggest priority actions based on current facility conditions, overcoming the static limitations of traditional textual inspection reports. A comparative analysis (Table 3) highlights the significant advantages of the proposed AEDT system over traditional manual inspections.

4.6 | Ablation Study

We conducted ablations to quantify each module's contribution (Table 4).

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Accuracy per class

Class	Accuracy	#Samples
Erosion-A	0.88	8
Erosion-B	1.00	15
Spalling-A	0.83	6
Spalling-B	0.88	8
Spalling-C	1.00	9
Siltation-A	0.83	6
Siltation-B	0.60	10
Siltation-C	0.77	13
Collapse-B	0.60	10
Collapse-C	0.94	16

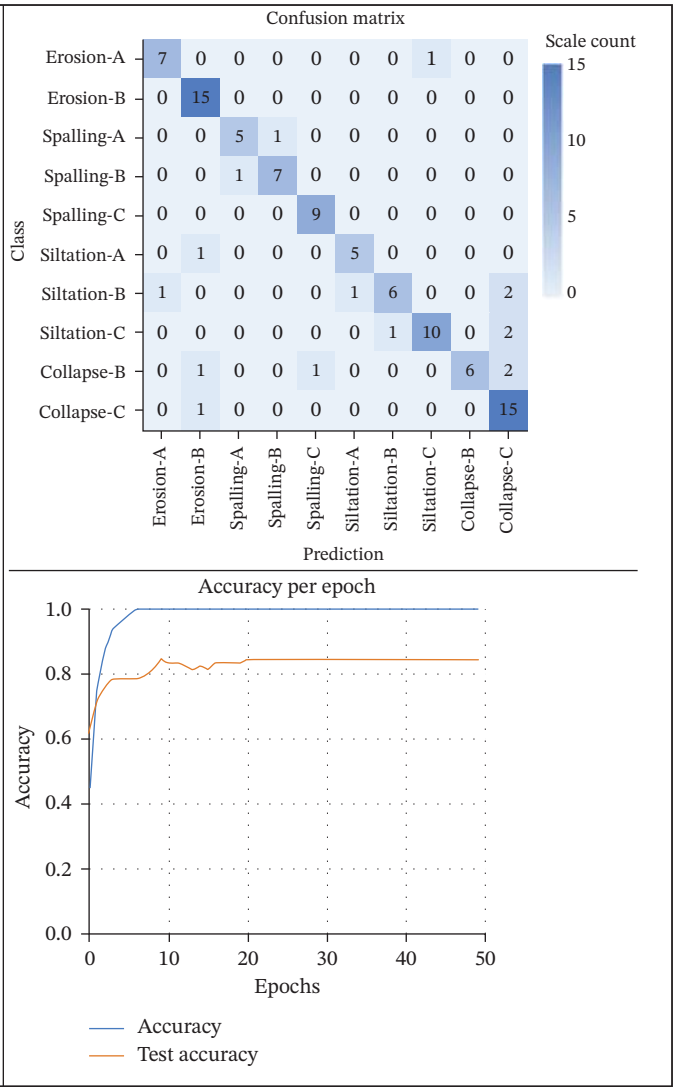


FIGURE 9 | Deep learning model testing.

TABLE 3 | Comparison between this system and traditional manual visual inspection.

Indicator	Traditional inspection	This system (DADT)
Average operation time	3.5 h	1.3 h
Subjective error	High	Low
Historical traceability	Paper-based records	Digital version records
Safety risk	High	Very low

TABLE 4 | Ablation results.

Configuration	Macro-F1	Notes
Full AEDT (proposed)	0.81	All modules enabled
(−) Data augmentation	0.72	Tests generalization without augmentation
(−) 3D context (2D reporting only)	N/A[a]	Qualitative user study on decision latency
(+) Traditional CV (SIFT + SVM)	0.58	Replaces CNN with classical pipeline

[a]3D context ablation compared via user-study metrics (task time, errors).

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These results demonstrate that data augmentation and CNN–based anomaly detection substantially contribute to overall performance, while 3D context primarily enhances decision efficiency rather than classification accuracy.

5 | Discussion

5.1 | Broader Implications and Benefits

Application results show that retaining structures—often rendered as surface elements—present visualization challenges due to low-texture realism. In contrast, river channel structures allow for more precise texture reproduction and were chosen for demonstrative modeling. The key benefits of the DT system in water conservation include:

- Design and planning: Enables early-stage simulation for enhanced design decision-making.
- Real-time monitoring: Uses IoT sensors for continuous facility condition tracking.
- Predictive maintenance: Anticipates maintenance needs based on degradation trends.
- Collaborative integration: Provides a shared interface for stakeholder communication.
- Disaster response: Supports early warnings and structured emergency workflows.
- Sustainability modeling: Simulates environmental impacts to inform long-term strategies.

In summary, the 3D DT developed in this project enhances water conservation inspection through semantic modeling, AI-driven defect detection, and immersive visualization. It reduces inspection time and costs, increases accuracy, and offers a scalable framework for future public infrastructure monitoring. These findings not only reinforce the technical feasibility of AI-enabled slope inspection but also contribute toward broader climate resilience strategies. By embedding DT technologies into environmental infrastructure systems, this research aligns with the goals of sustainable infrastructure (SDG 9) and climate adaptation (SDG 13), offering actionable insights for policy makers and field engineers managing landslide-prone regions.

Moreover, field interviews with maintenance engineers indicated that the AEDT system significantly streamlined documentation workflows, reducing the time spent on compiling visual inspection reports by over 40%. Additionally, the spatial tagging feature enabled direct correlation between observed defects and maintenance logs, facilitating cross-departmental coordination and audit compliance.

Specific use-case evaluations demonstrated that predictive maintenance was achievable through early identification of minor erosion patterns that historically preceded larger slope failures—allowing preemptive intervention weeks before visual symptoms intensified. In disaster response scenarios, the system's auto-generated reports and spatially calibrated inspection summaries could be exported within minutes in standardized formats, significantly accelerating emergency response coordination and reducing reliance on post-event manual surveys. In demonstration field sites, management authorities reported that the system effectively reduced on-site inspection labor requirements and assisted in prioritizing maintenance scheduling decisions based on spatial anomaly patterns and severity tags.

5.2 | Limitations of the Current Study

Despite its promising performance, the current system is constrained by reliance on RGB imagery, which limits detection of subsurface or visually occluded defects, such as those hidden by vegetation. Environmental disturbances like lighting variation or UAV motion artifacts also affect detection robustness. Future improvements may include the integration of multispectral or thermal imaging sensors to enhance the identification of partially obscured or concealed damage. Additionally, incorporating UAV stabilization modules and low-light enhancement algorithms could significantly improve image quality and detection reliability under challenging environmental conditions.

6 | Conclusion and Future Directions

This study developed an integrated inspection framework tailored for artificial slopes and SWC structures, combining UAV imaging, CNN, and semantic-enhanced SfM 3D reconstruction. The resulting system successfully achieved intelligent field-level inspections that significantly reduced manual labor demands while enhancing consistency and visualization of structural anomalies.

The proposed system consists of five modular components: (1) multiview UAV data acquisition; (2) SfM-based 3D model generation; (3) CNN–based anomaly pattern classification; (4) semantic embedding into a DT model; and (5) an interactive visualization and maintenance reporting interface. Field deployment at a water conservation infrastructure site demonstrated a classification accuracy of 81% for five distinct anomaly categories and reduced inspection time by over 60% compared to traditional methods.

A key contribution of this research is the introduction of the AEDT model, which integrates both geometric and semantic attributes—including damage type, severity, spatial distribution, and temporal change. This enriched digital representation enables enhanced maintenance scheduling, long-term monitoring, and risk-based prioritization, laying the foundation for intelligent asset management in hydrological engineering contexts.

Future work should focus on integrating LiDAR or multispectral imaging for structural penetration, as well as edge AI inference modules to facilitate real-time, in situ decision-making under diverse terrain and weather conditions. Ultimately, this research demonstrates the feasibility and value of embedding deep learning and DT technologies into slope infrastructure inspection workflows. It offers a scalable foundation for the broader adoption of smart monitoring systems in environmental risk management, infrastructure resilience, and climate adaptation planning.

Author Contributions

Nai-Hsin Pan: conceptualization, methodology, supervision, funding acquisition, writing – review and editing. **Bo-Yuan Chang**: data curation, formal analysis, investigation (UAV field operations and SfM

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reconstruction), writing – original draft. **Kuei-Yen Chen**: software (CNN model development and digital twin integration), validation, data curation (annotation and dataset management).

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available upon request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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